

Financial Statement Fraud Risk-Screening Model

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Summary

Internal control and audit teams can't review every filing, so the highest-leverage question is which ones to look at first. I built a risk-scoring model that ranks company filings by fraud likelihood using real SEC enforcement outcomes as ground truth, so a limited review team can prioritize the highest-risk cases instead of sampling at random. Scored out-of-time on 2002-2010 filings, the model's top 5% highest-risk flags contain 2.2x more of the eventual frauds than random sampling would, and the top 1% contain 3.6x more — concentrating scarce review capacity where it matters most.

Data source

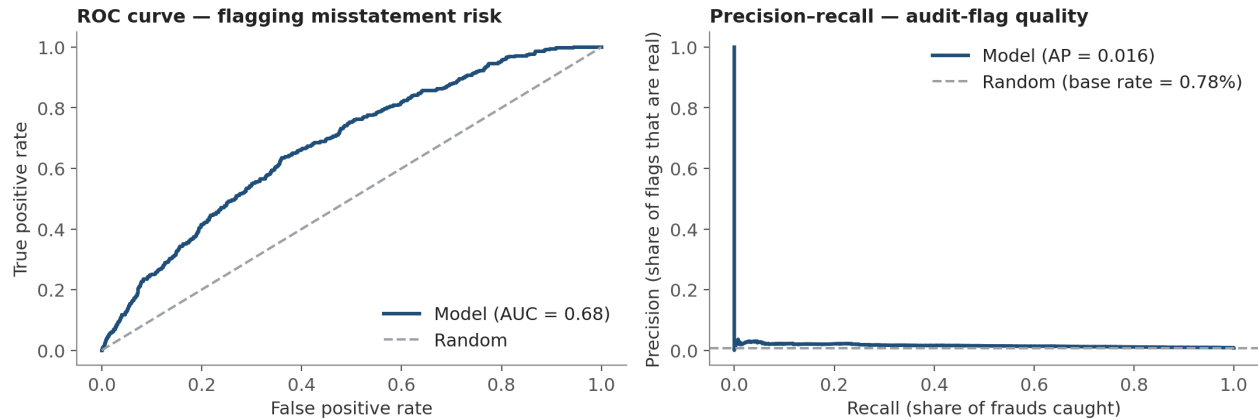
Bao, Ke, Li, Yu & Zhang (2020), “*Detecting Accounting Fraud in Publicly Traded U.S. Firms Using a Machine Learning Approach*,” *Journal of Accounting Research* 58(1): 199-235. The dataset pairs Compustat financial-statement data for 1991-2014 with fraud labels drawn from actual SEC Accounting and Auditing Enforcement Releases (AAERs) — i.e., companies the SEC formally charged with financial-statement fraud — giving 146,045 firm-year observations with a genuine (and highly imbalanced, ~0.7%) fraud rate. Publicly available at github.com/JarFraud/FraudDetection.

Methodology

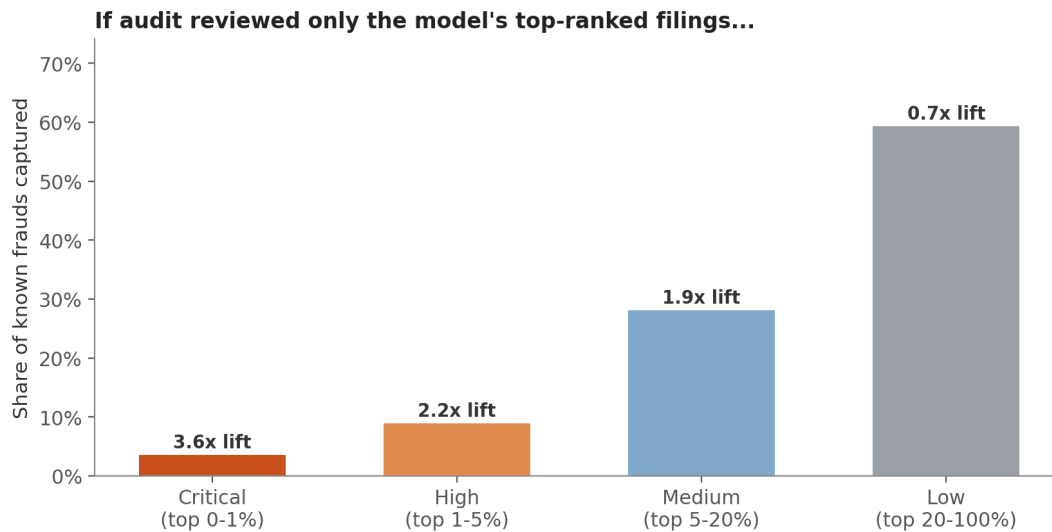
- **Data-quality control:** AAERs are typically filed years after the misstated fiscal year, so recent firm-years in any snapshot are under-labeled (frauds not yet charged look like clean filings). I excluded fiscal years after 2010 from evaluation to avoid crediting or penalizing the model on data that hadn't finished being labeled — the kind of data-integrity check a risk team has to make before trusting a metric.
- **Out-of-time validation:** trained on fiscal years 1991-2001, tested on unseen future years (2002-2010). A random train/test split would leak future information and overstate performance; this mirrors how a monitoring model is actually used — score filings you haven't seen yet.
- **Features:** 14 forensic-accounting ratios from the academic misstatement-prediction literature (e.g., change in receivables, change in free cash flow, soft-asset intensity, RSST accruals), rather than raw financials, so a one-unit change is comparable across companies of different sizes.
- **Models:** logistic regression with balanced class weights (interpretable coefficients double as risk-driver rankings) and a random forest as a non-linear robustness check, both evaluated with ROC-AUC, precision-recall AUC, and capture rate by review-priority tier — the metrics that matter under severe class imbalance, where plain accuracy is meaningless (99.3% accuracy from flagging nothing).

Results

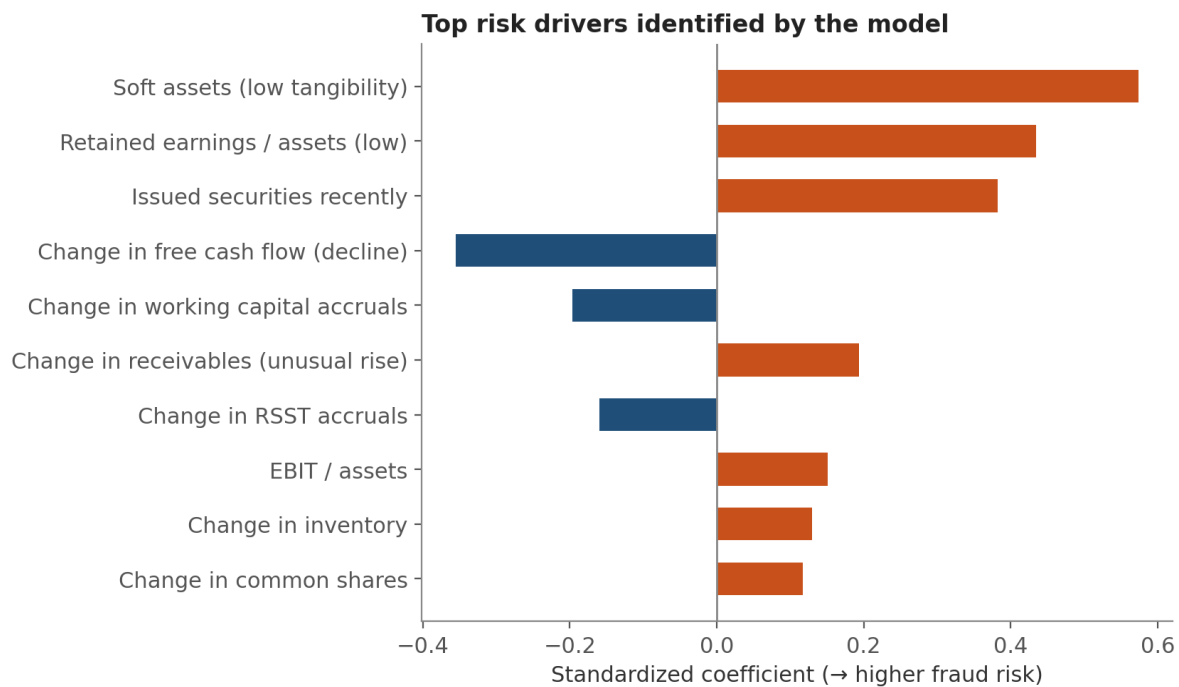
Out-of-time test (scored on 2002–2010 filings, model trained on 1991–2001)



The model separates fraud from non-fraud firm-years meaningfully better than chance (ROC-AUC 0.68, consistent with the range reported in the original study) despite fraud being a ~0.8% event in the test period — which is also why raw precision looks low in the PR curve: with so few positives, even a strong model produces mostly false positives in absolute terms. The tier view below is the more decision-relevant read.



If a review team could only examine the top 5% of filings by risk score, they'd catch about 9% of the actual frauds in the test set — 2.2x what a random 5% sample would catch. In the top 1%, the hit rate is 3.6x random. This is the practical case for a model like this: it doesn't replace human judgment, it tells a capacity-constrained team where to spend it first.



The strongest standardized risk drivers — higher soft-asset intensity (intangibles/goodwill-heavy balance sheets are harder to verify), low retained earnings relative to assets, recent securities issuance, and a declining change in free cash flow — line up with the classic red flags auditors are trained to probe: aggressive asset recognition, pressure to show growth after raising capital, and earnings that aren't backed by cash. That overlap with established forensic-accounting theory is itself a sanity check on the model, not just a modeling output.

Limitations & how I'd extend this

- Fraud is rare and the label depends on SEC enforcement being pursued and eventually filed — not every misstatement is caught, so the model is calibrated against *discovered* fraud, a known bound acknowledged in the source literature.
- Next step: layer in industry and auditor-level features, test a boosted-tree model matching the original paper's RUSBoost approach on the full 28 raw variables, and validate tier thresholds against a realistic review-capacity constraint (e.g., “we can only review 50 filings a quarter”) rather than a flat percentage.

Tools: Python (pandas, scikit-learn, matplotlib). Full code and data-prep pipeline available on request. Data: SEC AAER-labeled financial statements via Bao et al. (2020), Journal of Accounting Research.